

A Corrected Quadratic Payment for Forecast-Conditioned Optional Loads

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Abstract

Garjani, Shokri and Kebriaei propose a quadratic payment for efficient resource allocation, while Cai and coauthors schedule flexible demand using forecast nodal carbon intensity. We examine the displayed payment and a deliberately restricted use of that forecast. Q's strict uniqueness certificate conflicts with its implementation condition, and its payment can have multiple equilibrium prices and negative participant utility. An other-agent transfer corrects the equilibrium payment without changing best responses. For optional convex loads, a frozen nonnegative carbon forecast and a balanced carbon charge then implement a unique forecast-optimal allocation at every equilibrium, with equilibrium budget balance and nonnegative utility, provided a planner multiplier exists. A uniform counterfactual forecast-error premise yields an attributed-load regret bound; a separate accounting premise is needed for physical generation-emissions regret. These claims do not implement P's conserved workload or binary storage routing.

1 Setting and contribution

Q maximises the sum of strongly concave private valuations over compact convex local resource sets with componentwise aggregate capacity constraints. It proposes the quadratic payment in its Eq. (40), claiming a unique Nash equilibrium, efficient allocation, budget balance and individual rationality; it also studies stochastic learning. P forecasts day-ahead nodal carbon intensity (NCI) and schedules distributed data-centre (DDC) loads and mobile energy storage (MESS) against the frozen forecast in its Eq. (26). P's workload-balance Eq. (28) conserves service, and its MESS routing contains binary decisions. [3, 1]

Here the decision is an *optional* nonnegative bus-time load vector, with convex local sets containing zero. The forecast is a fixed coefficient in the agents' objective. This is a different allocation problem from P's mandatory DDC workload and mixed-integer MESS dispatch. The new claim concerns Q's displayed payment in this restricted problem, not all payments in Q's quadratic family.

Prior work already separates efficient allocations from unique equilibrium messages: Heydaribeni and Anastasopoulos exhibit a network mechanism with infinitely many Nash equilibria but the same efficient allocation, including multiplicity associated with dual prices [4]. Chen and Zhao already claim carbon-aware social optimality, budget balance and individual rationality in a joint electricity-carbon pricing mechanism [2]. Thus neither allocation-level implementation as a general pattern nor a balanced carbon charge alone

is the contribution here. The specific contribution is a diagnosis and equilibrium-payment repair for Q's Eq. (40), together with its limited forecast-conditioned consequence.

2 What fails for Q's displayed rule

Let $N \geq 2$, $K \geq 1$, $x_n \in \mathcal{X}_n \subseteq \mathbb{R}_+^K$, $p_n \in \mathbb{R}_+^K$, $c \in \mathbb{R}_+^K$, and $\alpha > 0$. Write t_n^Q for Q's Eq. (40), and set

$$\bar{p}_{-n} = \sum_{m \neq n} p_m, \quad \bar{x}_{-n} = \sum_{m \neq n} x_m, \quad S = \sum_n x_n - c.$$

For reference, the parts of Q's payment that depend on agent n 's own choice are

$$t_n^Q = \alpha \left[\frac{1}{2} p_n^\top p_n - \frac{p_n^\top (\bar{p}_{-n} + S)}{N-1} + \frac{N}{(N-1)^2} \bar{p}_{-n}^\top x_n \right] + \text{terms independent of } (x_n, p_n).$$

This expression is only a best-response representation; equilibrium payment levels below use the complete Eq. (40).

Proposition 1 (Failure of the strict LMI certificate). *Q's Proposition 1 condition $\Psi + \Psi^\top \leq -\nu I$, $\nu > 0$, is incompatible with its Theorem P2(i) condition $\sum_m A_{nm}^n = 0$ for every n . This rules out their joint use as a certificate, without ruling out efficient implementation.*

Proof. Take a nonzero $q \in \mathbb{R}^K$ and a direction ν whose allocation blocks are zero and whose price blocks all equal q . Q's price-price blocks give

$$\nu^\top \Psi \nu = - \sum_n q^\top \left(\sum_m A_{nm}^n \right) q = 0.$$

Hence $\nu^\top (\Psi + \Psi^\top) \nu = 0$, contrary to strict negative definiteness. $\square$

Full strategy uniqueness fails directly for Eq. (40). Let $N = 2$, $K = 1$, $\alpha = 1$, $c = 2$, $\mathcal{X}_1 = \mathcal{X}_2 = [0, 1]$, and $V_n(x) = 3x - x^2/2$. Every $(x_1, x_2, p_1, p_2) = (1, 1, p, p)$, $0 \leq p \leq 2$, is an equilibrium: a unilateral move $(x_n, p_n) = (1-d, p+u)$ changes utility by $-(2-p)d - (d+u)^2/2 \leq 0$. The efficient allocation is unique, while the capacity multiplier and equilibrium price are not.

The same payment need not be individually rational. For $N = 2$, $\alpha = c = 1$, $\mathcal{X}_n = [0, 1]$, $V_1(x) = 3x - x^2/2$, and $V_2(x) = 2x - x^2/2$, the profile $(x_1, x_2, p_1, p_2) = (1, 0, 2, 2)$ is an equilibrium. Q's payments are $(3, -3)$, so player 1 receives $V_1(1) - 3 = -1/2 < 0 = V_1(0)$. These examples address Q's displayed Eq. (40) and its associated claims.

3 A restricted correction

Assume each $\mathcal{X}_n$ is compact and convex, $0 \in \mathcal{X}_n \subseteq \mathbb{R}_+^K$, and each V_n is continuously differentiable, α -strongly concave and satisfies $V_n(0) = 0$. Let $\hat{e} \in \mathbb{R}_+^K$ be frozen, $\beta \geq 0$, and assume the planner below has a primal-dual KKT pair. These are assumptions for the optional-load result, rather than properties established for P's full dispatch.

$$\max_{x_n \in \mathcal{X}_n, \sum_n x_n \leq c} F_{\hat{e}}(x), \quad F_{\hat{e}}(x) = \sum_n V_n(x_n) - \beta \hat{e}^\top \sum_n x_n. \quad (1)$$

Define two additions to Q's displayed payment:

$$h_n = \frac{\alpha N}{(N-1)^2} \left(\frac{\bar{p}_{-n}}{N-1} \right)^\top \left(\bar{x}_{-n} - \frac{N-1}{N} c \right), \quad (1)$$

$$b_n = \beta \hat{e}^\top x_n - \frac{\beta}{N-1} \hat{e}^\top \bar{x}_{-n}, \quad t_n = t_n^Q + h_n + b_n. \quad (2)$$

The transfer h_n uses only other agents' actions. The second term of b_n also uses only other agents' actions, while $\sum_n b_n = 0$ at every strategy profile.

Theorem 1 (Forecast-conditioned allocation and settlement). *Under the assumptions above, the game with payments (1)–(2) has an equilibrium. Every equilibrium has the unique allocation solving (1); equilibrium prices may differ across equilibria. At every equilibrium the payments sum to zero and every agent's utility is nonnegative relative to the optional zero-service benchmark.*

Proof. Both additions leave the price best response unchanged; h_n leaves every best response unchanged, and b_n replaces V_n by the still strongly concave $\tilde{V}_n = V_n - \beta \hat{e}^\top x_n$. For coordinate k , write $P^k = \sum_n p_n^k$. Q's displayed rule yields

$$p_n^k = \left[\frac{P^k - p_n^k + S^k}{N-1} \right]_+.$$

If any $p_n^k > 0$, then $Np_n^k = P^k + S^k > 0$, forcing every agent's price in that coordinate to be positive. They are equal, and summation gives $S^k = 0$. If all prices are zero, $S^k \leq 0$. Thus prices are common, $p \geq 0$, allocations are feasible, and $p^\top S = 0$.

At common prices the marginal payment for an agent's allocation is αp . Allocation best responses are precisely the planner's local KKT conditions with multiplier $\lambda = \alpha p$. Strong concavity gives a unique primal allocation. Conversely, a planner KKT pair supplies a common-price candidate $p_n = \lambda/\alpha$; each player's payoff is jointly concave in its own allocation and price under the stated strong concavity and $N \geq 2$, so the KKT conditions make this candidate an equilibrium.

At equilibrium, direct substitution into the complete Q payment and complementarity give

$$t_n^Q = \alpha \left(1 + \frac{N}{(N-1)^2} \right) p^\top (x_n - c/N), \quad t_n^Q + h_n = \alpha p^\top (x_n - c/N).$$

These corrected base payments sum to zero by complementarity, and the carbon additions sum to zero identically. Finally, the allocation best response against feasible $x_n = 0$ gives $V_n(x_n) - \beta \hat{e}^\top x_n - \alpha p^\top x_n \geq 0$. Agent n 's equilibrium utility is this quantity plus $\alpha p^\top c/N + \beta \hat{e}^\top \bar{x}_{-n}/(N-1)$, which is nonnegative. The budget claim for h_n is at equilibrium. $\square$

4 Forecast error and its interpretation

Let $\mathcal{F}$ be the feasible allocations of (1), $X(x) = \sum_n x_n$, and let $e(x)$ denote a possibly dispatch-dependent realised NCI. Define attributed-load welfare $F_e(x) = \sum_n V_n(x_n) - \beta e(x)^\top X(x)$. This is an accounting objective; P's calculated NCI in Eq. (23) and fixed-forecast objective in Eq. (26) do not identify it with changes in physical generation emissions.

Proposition 2 (Conditional regret). *If $\sup_{x \in \mathcal{F}} \|e(x) - \hat{e}\|_\infty \leq \varepsilon$, then the forecast-optimal equilibrium allocation $\hat{x}$ has attributed-load welfare regret at most $2\beta\varepsilon\|c\|_1$. If realised intensity is instead a fixed vector e with $\|e - \hat{e}\|_\infty \leq \varepsilon$, the bound is $\beta\varepsilon\|c\|_1$. Neither premise follows from P's reported observational forecast errors.*

Proof. For each feasible x , $0 \leq X(x) \leq c$, so $|F_e(x) - F_{\hat{e}}(x)| \leq \beta \epsilon \|c\|_1$. Compare a maximiser of F_e and $\hat{x}$ on both sides of forecast optimality. For a fixed e , the forecast error instead multiplies $X(x_e) - X(\hat{x})$, whose ℓ_1 norm is at most $\|c\|_1$. $\square$

If actual generation emissions are $C(x)$ and a fixed baseline is C_0 , the additional uniform accounting link $\sup_{x \in \mathcal{F}} |C(x) - C_0 - e(x)^\top X(x)| \leq \delta$ gives regret at most $2\beta(\epsilon \|c\|_1 + \delta)$ for welfare $\sum_n V_n(x_n) - \beta(C(x) - C_0)$. This follows by comparing physical welfare with forecast welfare at both feasible allocations. P establishes neither uniform premise.

To see why this qualification matters, consider a hypothetical two-location dispatch. Zero-carbon generation at A already meets its baseline demand and an added MWh there uses backup generation emitting one unit. At B, the added MWh emits one-half unit. Baseline average NCI is zero at A and one-half at B, so a frozen-NCI objective ranks A first even though B has lower incremental physical emissions. This is an interpretation stress test, not a result from P's IEEE 33-bus simulation.

5 Limitations and open questions

The theorem changes P's service model: DDC workload is conserved in P, and MESS movement is binary. A zero-service outside option and Q's componentwise capacity form therefore do not establish participation or implementation for P's exact programme. The correction also leaves price multiplicity possible and supplies no stochastic-learning convergence result. Its equilibrium budget balance does not assert that h_n balances the budget off equilibrium.

The forecast regret bound requires a uniform *counterfactual* error guarantee over feasible decisions. A physical-emissions conclusion additionally requires an accounting link to generation emissions. A concrete next question is to specify a small dispatch with an explicit $C(x)$, derive its counterfactual $e(x)$, and test both bounds. The present analysis does not establish those premises or an empirical improvement in P's network.

References

- [1] Feiyu Cai, Jing Qiu, Yi Yang, Chenxi Zhang, Xinlei Wang, Baichuan Liu, and Junhua Zhao. Emission-forecasting-based spatial-temporal carbon response: A multi-agent attention-enhanced deep learning framework, 2026. URL: <https://arxiv.org/abs/2607.26560>, [arXiv:2607.26560](https://arxiv.org/abs/2607.26560).
- [2] Yue Chen and Changhong Zhao. Achieving social optimum and budget balance via a joint electricity-carbon pricing mechanism, 2024. URL: <https://arxiv.org/abs/2308.08195>, [arXiv:2308.08195](https://arxiv.org/abs/2308.08195).
- [3] Shaya Garjani, Mohammad Shokri, and Hamed Kebriaei. A systematic approach to mechanism design with stochastic dynamic stability, 2026. URL: <https://arxiv.org/abs/2608.29130>, [arXiv:2608.29130](https://arxiv.org/abs/2608.29130).
- [4] Nasimeh Heydaribeni and Achilleas Anastasopoulos. Distributed mechanism design for network resource allocation problems, 2019. URL: <https://arxiv.org/abs/1904.01222>, [arXiv:1904.01222](https://arxiv.org/abs/1904.01222).